F172 Lesson 11: Genetic Engineering vs Gene Therapy

OCR AAQ Level 3 Human Biology · Unit F172: Genetics (Matt)

How does industrial genetic engineering fundamentally differ from human clinical gene therapy?

- Core Definitions
- Recombinant Tools
- Somatic vs Germline
- Ethical & Legal Governance

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Genetic Engineering: Broad Biotechnology & Transgenics



Definition

- **Artificial manipulation:** Direct alteration of an organism's genome using biotechnology.
- **Novel synthesis:** Modifying biological characteristics to synthesize entirely new products.

Recombinant Tools

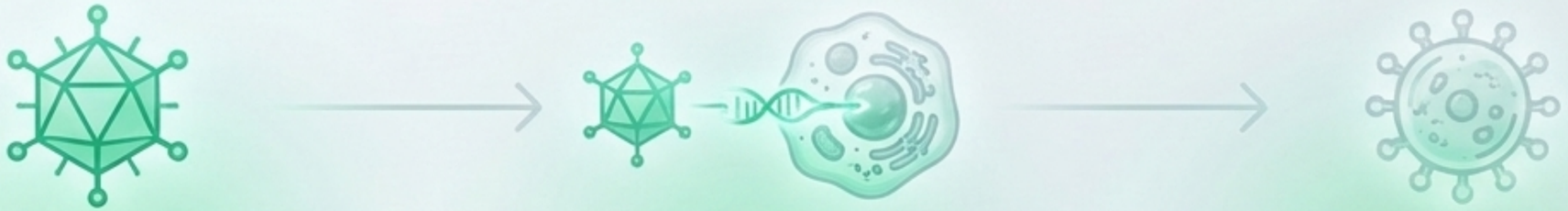
- **Gene isolation:** Extracting specific DNA sequences of interest.
- **Plasmid insertion:** Utilizing restriction enzymes and DNA ligase to build vectors.
- **Host transformation:** Introducing recombinant DNA into non-human host cells.

Commercial Niches

- **Transgenic bacteria:** Mass-producing human insulin and clotting Factor VIII.
- **Modified crops:** Engineering agricultural staples like Bt maize and golden rice.
- **Animal models:** Creating transgenic organisms strictly for research purposes.

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Gene Therapy: Targeted Human Clinical Medicine



Medical Definition

- **Targeted alteration:** Introduction, removal, or modification of human genetic material.
- **Clinical intent:** Specifically designed to treat, prevent, or cure human disease.

Clinical Purpose

- **Genetic root:** Correcting monogenic pathology precisely at its source.
- **Beyond symptoms:** Moving from downstream symptom management to physiological restoration.

Targeted Disorders

- **Monogenic conditions:** Severe inherited disorders like cystic fibrosis and SCID.
- **Neurodegeneration:** Targeting complex adult-onset diseases like Huntington's.
- **Acquired diseases:** Advancing precision oncology and cancer immunotherapies.

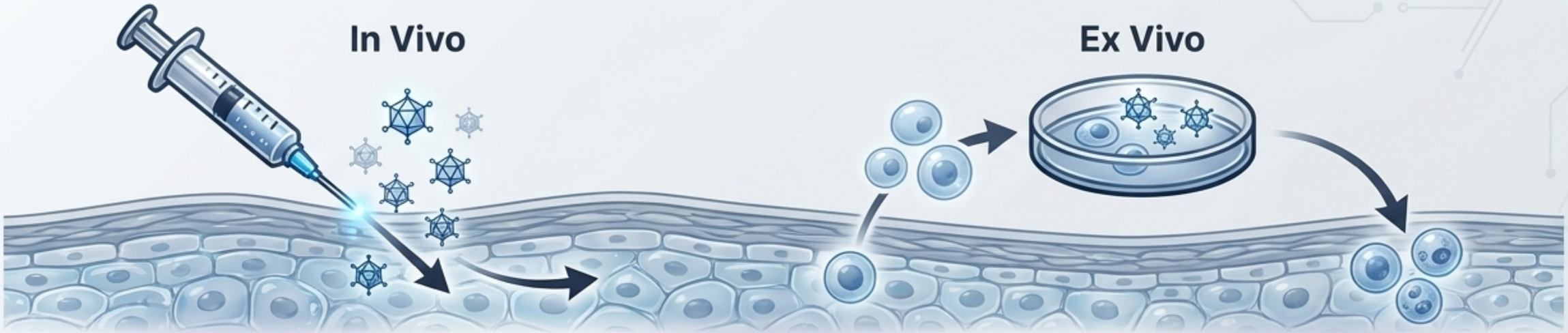
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Comparative Matrix: Engineering vs Gene Therapy

	Genetic Engineering	Gene Therapy
Primary Objective	Industrial synthesis and agricultural improvement.	Alleviating human disease and restoring normal physiological function.
Target Organisms	Bacteria, plants, fungi, and non-human animals.	Exclusively human patients.
Delivery System	Plasmids, Agrobacterium vectors, and microinjection.	Engineered viral vectors (AAV, Lentivirus) and targeted Lipid Nanoparticles (LNPs).
Regulatory Arena	Agricultural and environmental biotechnology oversight boards.	Human clinical trial regulatory agencies (MHRA, FDA).

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Biological Delivery: Vectors & Pathways



Delivery Pathways

- **In Vivo:** Corrective vectors injected directly into specific patient tissues.
- **Ex Vivo:** Patient cells extracted, genetically modified in vitro, and reimplanted.
- **In Situ:** Directly targeting a highly localized cellular environment.

Viral Vectors

- **Adeno-Associated Virus (AAV):** Engineered viruses safely stripped of pathogenicity.
- **Targeted tropism:** Delivering therapeutic transgenes specifically to neural or muscle cells.

Non-Viral Systems

- **Lipid Nanoparticles (LNPs):** Chemically encapsulating genetic material for safe cellular transit.
- **CRISPR-Cas9:** Precision ribonucleoprotein complexes for targeted genome repair.

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Somatic Cell Gene Therapy: The Clinical Standard



Target Cells

- **Differentiated tissues**: Targeting non-reproductive adult or paediatric somatic cells.
- **Specific sites**: Intervening in bone marrow, retinal pigment epithelium, or striatal neurons.

Heritability

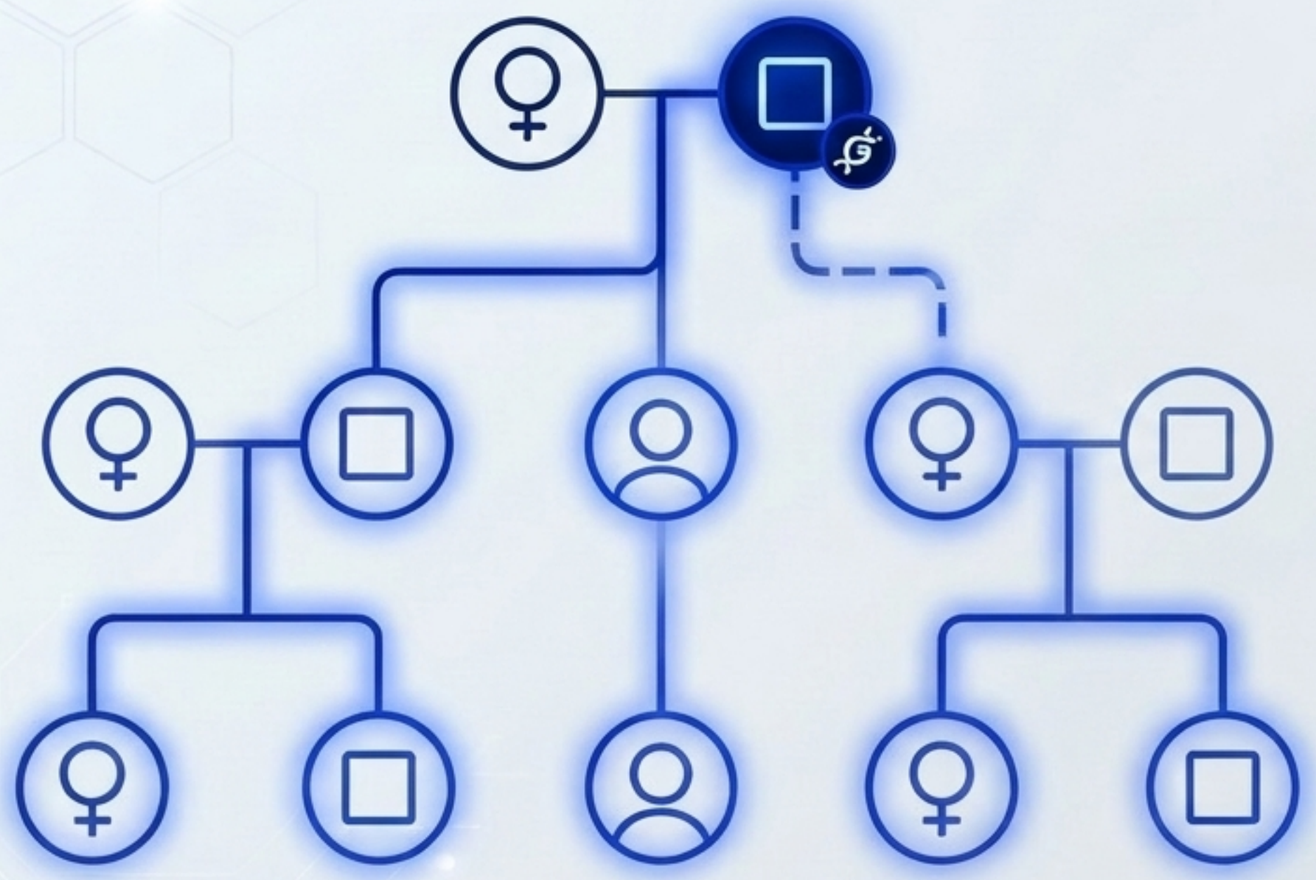
- **Strictly non-heritable**: Modifications affect only the treated individual patient.
- **Biological dead-end**: Genetic changes perish completely with the treated somatic cells.

Approved Therapies

- **Zolgensma**: Intravenous AAV therapy for Spinal Muscular Atrophy (£1.4m single-dose).
- **Luxturna**: In vivo retinal injection reversing specific inherited visual dystrophies.
- **Casgevy**: Ex vivo CRISPR therapy curing severe sickle cell disease.

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Germline Gene Therapy: The Generational Frontier



Target Cells

- **Reproductive cells:** Directly modifying human spermatozoa or oocytes.
- **Early embryos:** Altering pre-implantation IVF embryos before cellular differentiation.

Heritability

- **Permanent transmission:** Modifications incorporate into every resulting cell lineage.
- **Ancestral cascade:** Genetic alterations are passed indefinitely to all future generations.

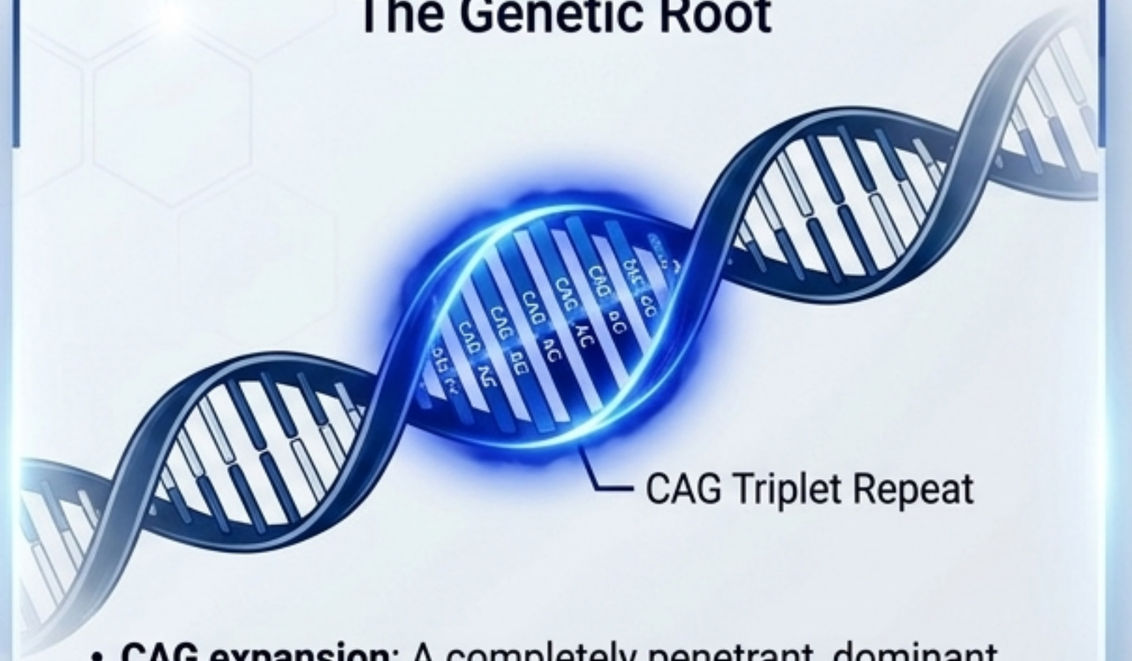
Biological Hazards

- **Off-target mutations:** Unintended genetic errors irreversibly propagated down the lineage.
- **Epigenetic unpredictability:** Disrupting highly complex, delicate embryonic developmental processes.

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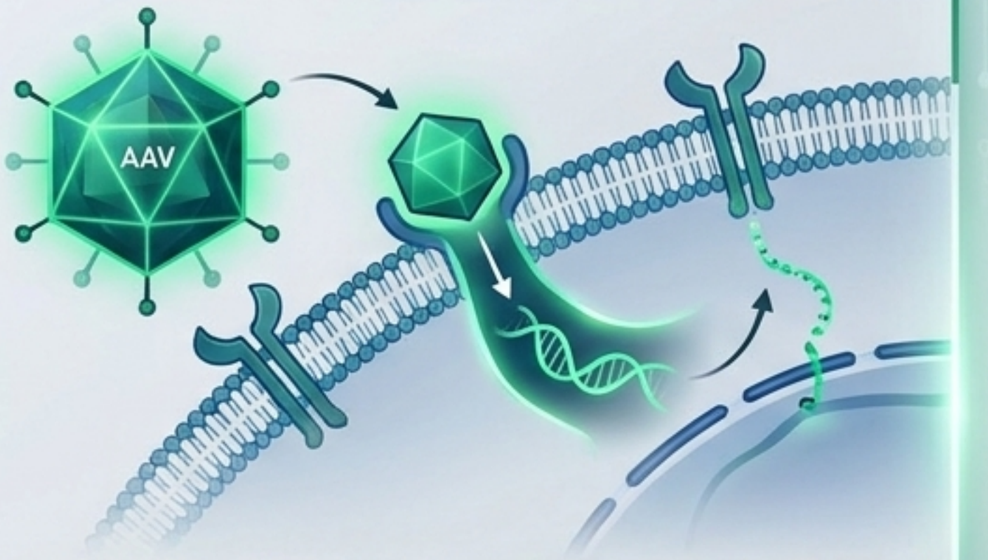
Clinical Focus: Targeting Huntington's Disease

The Genetic Root



- **CAG expansion**: A completely penetrant, dominant mutation in the IT15 (HTT) gene.
- **Cellular toxicity**: Mutant huntingtin (mHTT) protein relentlessly destroys the corpus striatum.
- **Adult onset**: Progressive motor deterioration typically beginning between 21 and 50 years.

The Therapeutic Frontier



- **Predictive testing**: Identifying the genetic defect decades before progressive symptoms manifest.
- **AMT-130 Trials**: uniQure's AAV-delivered therapy attempting to silence toxic mHTT expression.
- **AB-1001 Therapy**: Gene transfer designed to increase CYP46A1 and restore brain cholesterol metabolism.

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Ethical, Societal & Distributive Dilemmas

Privacy & Consent

- **Consent impossibility:** Future generations cannot consent to germline modifications.
- **DNA Databases:** Storing genetic data creates profound risks of surveillance and discrimination.

Genetic Enhancement

- **The slippery slope:** Shifting from curing monogenic pathology to selecting non-medical traits.
- **Designer babies:** Amplifying existing societal inequality through engineered genetic advantages.

Healthcare Equity

- **Astronomical pricing:** Single-dose curative therapies currently cost between £1.4m and £2.8m.
- **Global rationing:** Creating severe domestic and international healthcare access divides.

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Regulatory Oversight: The UK Gold Standard



HFEA Authority (Embryology)

- **Embryo research**: Strict 14-day legal limits on human embryo manipulation.
- **Germline prohibition**: Clinical germline gene editing remains universally illegal in the UK.

MHRA Governance (Medicines)

- **Clinical trials**: Rigorous, multi-phase oversight of human gene therapy
- **Somatic approval**: Regulating modern gene therapies strictly as advanced medicinal products.

- **Clinical trials**: Rigorous, multi-phase oversight of human gene therapy safety.

The Balancing Act (Society)

- **Protecting society**: Managing biotechnological power with absolute ethical guardrails.
- **Enabling innovation**: Supporting safe, equitable avenues for curing severe genetic disorders.

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